

PROPOSAL PACKAGE | DOCUMENT 06

Technology Stack and Tools

Every software product and service proposed for the project, what it does, why it was chosen, what it costs, and the alternatives

Document	06 of 15
Audience	Client IT, procurement, delivery team; also a reference for the presenter
Version	1.0 (first-meeting edition)
Date	September 2026
Classification	Confidential - prepared for the client

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1. Selection principles

- **Open standards and open source where it matters for portability:** PostgreSQL, containers, Terraform, OpenTelemetry. The client can move providers.
- **Managed services during pilot to keep cost and operations minimal,** replaced by in-region or self-hosted equivalents at enterprise tier without changing application code.
- **Best-in-class AI components, but always behind the gateway** so no single vendor is load-bearing.
- **Arabic and English support is a hard requirement** for every component that touches text.
- Prices below are list prices as published in 2026 and are indicative; they change, and enterprise agreements differ. The Cost and Pricing Model workbook (Document 05) holds the live numbers.

2. Stack at a glance

Layer	Pilot choice	Enterprise / sovereign choice
Web front end	Next.js on Vercel	Next.js in containers behind the client's load balancer
Core API	Node.js (TypeScript)	Same, containerised
Intelligence service	Python, FastAPI	Same, containerised, GPU node pool for self-hosted models
Database	PostgreSQL with pgvector (Supabase)	Managed PostgreSQL in region (AWS RDS, Azure Database for PostgreSQL) or self-managed
Object storage	Supabase Storage	S3 / Azure Blob in region, or MinIO on premises
Queue	PostgreSQL-backed queue	Redis (managed) with workers
Identity	Supabase Auth with OIDC to client IdP	Microsoft Entra ID / Okta; Keycloak if self-hosted
Model gateway	LiteLLM	LiteLLM (high availability)
Language models	Claude Opus 5 and Claude Sonnet 5 via Anthropic API or AWS Bedrock	Same via Bedrock / Foundry in region, plus Qwen 3.5 / GLM-5 / Gemma 4 on vLLM for sovereign routing
Embeddings	Voyage AI voyage-3.5	Same, or BGE-M3 self-hosted
Re-ranking	Voyage rerank-2.5	Same, or bge-reranker-v2-m3 self-hosted
Document parsing	Docling (open source)	Docling
OCR (scans, Arabic)	Azure AI Document Intelligence (UAE North)	Same, or Tesseract / PaddleOCR self-hosted
Observability	Langfuse, Sentry	Langfuse self-hosted, Grafana stack, SIEM export
Infrastructure as code	Terraform	Terraform, policy as code
CI/CD	GitHub Actions	GitHub Actions or Azure DevOps
Security tooling	ClamAV, cloud KMS, Cloudflare	Same plus HSM-backed keys, SIEM

Layer	Pilot choice	Enterprise / sovereign choice
	or cloud WAF	

3. Component by component

3.1 Language models (the "reasoning engine")

Item	Detail
What it does	Reads retrieved passages and writes answers; extracts claims into structured rows; adjudicates whether two textual terms conflict; checks that each answer sentence is supported; drafts outreach text on request.
Primary choice	Claude Opus 5 (model id claude-opus-5) for answer synthesis, contradiction adjudication and grounding checks; Claude Sonnet 5 (claude-sonnet-5) for bulk claim extraction and routine lookups. Both have a 1 million token context window, which lets the extraction pass read long contracts whole.
List price (Anthropic API, 2026)	Opus 5: USD 5 per million input tokens, USD 25 per million output tokens. Sonnet 5: USD 2 / USD 10. Haiku 4.5 (USD 1 / USD 5) is available for simple classification steps. Prompt caching and batch processing cut ingestion costs materially (batch is 50% off).
Access routes	Anthropic API direct (pilot); Amazon Bedrock from the AWS Middle East regions (UAE me-central-1, Bahrain me-south-1) using global cross-region inference; Microsoft Foundry (billed through the Microsoft marketplace). Route is a gateway setting.
Why	Strong performance on long, structured documents and on instruction-following for citations and refusals; enterprise terms; multiple GCC-reachable access routes.
Alternatives	OpenAI GPT-5 family via Azure OpenAI (also reachable in UAE North); Google Gemini via Vertex AI. All are supported by the gateway; a switch is configuration.
Sovereign option	Open-weight models served in region with vLLM: Qwen 3.5, GLM-5, Gemma 4, DeepSeek V4 (licence review needed per model). Quality is evaluated against the golden set before switching.

3.2 Model gateway: LiteLLM

Open-source proxy that exposes one internal endpoint for all model calls. It gives per-team keys, spend limits, request logging, retries, fallbacks and **routing rules by tag** (for example: documents classified "restricted" go only to the in-region model). It removes vendor lock-in and makes the residency pattern a configuration change. Free to self-host; enterprise support available.

3.3 Embeddings and re-ranking

Item	Detail
What they do	Embeddings turn text into vectors so meaning-based search works across languages ("contract value" finds "قيمة العقد"). The re-ranker reads the question and each candidate passage together and orders them by relevance, which sharply improves the quality of

Item	Detail
	what the model sees.
Choice	Voyage AI voyage-3.5 (multilingual; USD 0.02 per million tokens; 200 million free tokens per account) and Voyage rerank-2.5. Voyage is now part of MongoDB; the API remains standalone.
Alternatives	Cohere Embed v4 and Rerank 3.5 (multilingual, strong Arabic); OpenAI text-embedding-3-large; self-hosted BGE-M3 and bge-reranker-v2-m3 (open source, multilingual, no data leaves the environment).

3.4 Document parsing and OCR

Item	Detail
Docling	Open-source parser (IBM) that converts PDF, DOCX, PPTX, XLSX and HTML into structured text with tables and reading order preserved; runs locally, no data leaves the environment; free.
Azure AI Document Intelligence	Cloud OCR and layout analysis with Arabic support, available in Azure UAE North for in-region processing. Priced per page (roughly USD 1.50 per 1,000 pages for read/layout tiers; check current list). Used only for scanned pages.
Alternatives	Google Document AI; AWS Textract (limited Arabic); self-hosted Tesseract or PaddleOCR (free, lower accuracy on poor scans); LlamaParse or Unstructured (managed parsing APIs).

3.5 Database: PostgreSQL with pgvector

One database for records, full-text search and vectors. pgvector provides approximate nearest-neighbour search (HNSW) sufficient to millions of chunks; PostgreSQL full-text search handles exact numbers, names and Arabic terms; row-level security enforces per-user access inside the database. Supabase provides a managed PostgreSQL with storage and auth for the pilot (**Pro plan USD 25 per month plus usage**); enterprise tier uses the client's cloud provider's managed PostgreSQL in region. If the corpus grows past what pgvector handles comfortably, Qdrant or OpenSearch can be added behind the same retrieval interface.

3.6 Web application and hosting

Next.js (React, TypeScript) for a fast, accessible, right-to-left-capable interface. Vercel hosts it during pilot (**Pro plan USD 20 per seat per month**); at enterprise tier the same application runs in containers inside the client's cloud. Tailwind CSS and a component library for consistent design; Playwright for end-to-end tests.

3.7 Identity and access

Single sign-on through the client's identity provider (Microsoft Entra ID is the norm in the GCC; Okta, Google Workspace also supported) using OpenID Connect. Multi-factor authentication is enforced by the provider. Roles and per-document access are managed in the application. Keycloak is the self-hosted alternative for the sovereign pattern.

3.8 Messaging channels for approved outreach

Channel	Tool
Email, Microsoft 365	Microsoft Graph API (send-as the approver, replies land in the normal mailbox); requires admin consent
Email, Google Workspace	Gmail API
WhatsApp	WhatsApp Business Platform (Meta Cloud API) with approved message templates; per-conversation pricing; strongly used in GCC relationship management
SMS	Twilio or a regional provider (Unifonic)
Internal reminders	Email digest, Microsoft Teams or Slack webhook, in-app

3.9 Observability and evaluation

- **Langfuse** (open source): traces every AI call with prompt, retrieved passages, answer, tokens and cost; supports evaluation datasets and scoring. Self-hostable, so traces stay in the environment.
- **OpenTelemetry + Grafana or Sentry**: application metrics, logs, errors, alerts.
- **Promptfoo / Ragas**: evaluation harness for the golden question set; runs in CI.

3.10 Infrastructure, security and delivery tooling

Purpose	Tool
Infrastructure as code	Terraform (HashiCorp), modules per environment
Containers	Docker; AWS ECS/Fargate, Azure Container Apps, or Kubernetes at scale
CI/CD	GitHub Actions: unit tests, evaluation gate, dependency and container scanning, deploy
Secrets	AWS Secrets Manager / Azure Key Vault; no secrets in code
Anti-malware	ClamAV in the ingestion path
Edge protection	Cloudflare WAF or the cloud provider's WAF; rate limiting
Backups	Provider-managed point-in-time recovery; cross-region copies
Source control and work tracking	GitHub (client-owned organisation), Linear or Jira, Notion or Confluence
Design	Figma

4. Licensing and ownership

- All custom code is delivered under the client's ownership (see Document 14).
- Open-source components are used under permissive or copyleft licences that permit internal use without disclosure (PostgreSQL, Apache-2.0, MIT). Where a component is AGPL (for example

some CRM products we considered but did not select), we note it; our custom CRM avoids AGPL dependencies.

- Commercial services (Anthropic, AWS, Azure, Voyage, Vercel, Supabase, Meta WhatsApp) are contracted in the client's name so the client controls the relationship and the data.

5. What we considered and did not choose for the core

Option	Reason not chosen as the core
Off-the-shelf CRM (HubSpot Starter USD 20 per seat per month; Attio ~USD 34; Salesforce) as the System 2 core	Strong products, but the mandatory human-approval gate, GCC pipeline semantics, in-region hosting and the shared entity model with System 1 would all be customisations on someone else's platform, with per-seat fees and data outside the GCC. Remains a valid choice if the client later wants a 500-seat sales organisation; our data model exports cleanly to them.
Twenty CRM (open source) as the System 2 core	Attractive and self-hostable, but AGPL-licensed and still thin on approval workflow and reporting; we may reuse its interface patterns.
Microsoft Copilot / SharePoint Q&A	Useful for general search, but it cannot enforce closed-domain answers with a grounding check, has no claims ledger, and cannot express decision checklists.
LangChain / LlamaIndex as the application framework	Helpful libraries; we use them selectively for parsers and evaluation, but keep the pipeline code explicit so behaviour is inspectable and auditable.
Dedicated vector database from day one (Pinecone, Weaviate)	Not needed at pilot scale; pgvector keeps one system of record. Qdrant is the upgrade path if needed.